KM CODE CRAFT

Radiology information system · Image archive · Reporting

Your whole radiology department in one system. On your premises.

From registration to the appointment, the modality and the released report: one workplace in the browser, in Arabic, English and German. The server and the data stay with you, not in a cloud.

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01 At a glance

Scope	RIS, PACS archive, web viewer, reporting, documents	Operation	Your own on-site server, works without internet
Languages	Arabic (RTL), English, German	Modalities	DICOM receive and modality worklist
Access	11 roles, second factor, LDAP/AD optional	Status	Version 0.1.0 · pilot phase

Where we honestly stand

The system is in its pilot phase. Images in the built-in viewer are not released for primary diagnosis; the program says so on every page. It is not approved as a medical device, and what approval is needed depends on the country you use it in. We are looking for sites for pilot installations.

02 What the system does

Every module can be switched on separately. You start with what you need.

Patient record

Demographics with a complete correction history, search by name and number. Duplicate records are merged under a two-person rule and can be separated again.

Orders and worklist

Create and follow examinations: requested, scheduled, arrived, in progress, completed. Each role sees only the steps it may perform.

Scheduling

Rooms and devices with protection against double booking, check-in on the day, cancellation and no-show.

Worklist at the modality

Scheduled examinations appear on the device through DICOM modality worklist. Staff pick the patient instead of typing; mix-ups from typing errors disappear.

Image archive (PACS)

DICOM receive with automatic matching of unambiguous studies to their order. Anything doubtful goes to a review list and is assigned or rejected by a person.

Image viewing

Web viewer straight from the examination through a short-lived personal link. For review and demonstration, not for primary diagnosis.

Reporting

Draft with templates, submit, release by a second reader (can be switched off), return with a reason, addenda. A released report cannot be changed.

Seal and PDF/A

Every released report receives a cryptographic system seal and can be exported as PDF/A-2b for long-term filing.

Documents

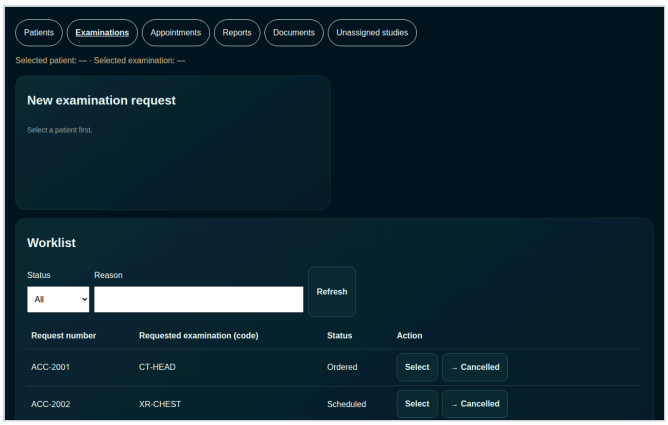
Upload referrals, consent forms and previous reports as PDF or image. Every file is scanned for malware before it is stored.

Administration

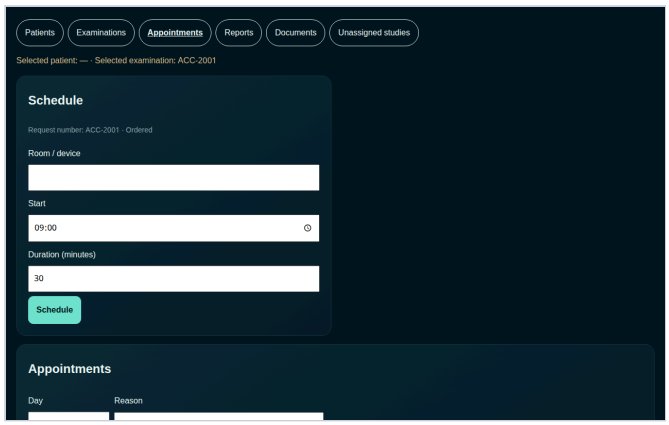
Overview of system state, waiting work and enabled functions. Users, roles and second factor in a dedicated administration console.

03 What the work looks like

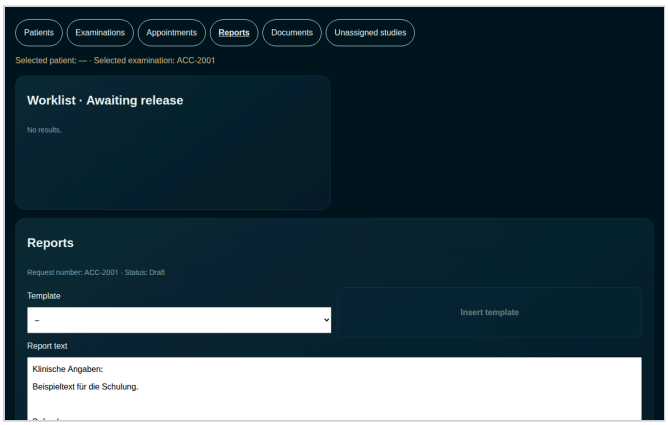
Screenshots of the real interface with invented sample data.



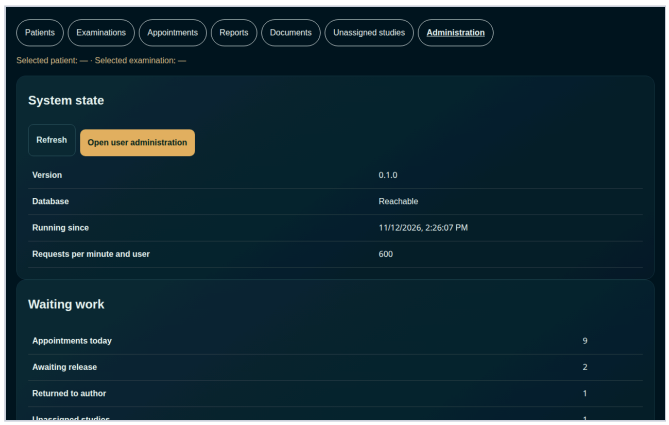
Worklist: examinations with status and the next permitted steps



Schedule: room, start, duration – double booking is prevented by the server



Reporting: draft, template, submit for release



Administration: state, waiting work, enabled functions

04 The right role for every task

Eleven roles separate who may see and do what. The server enforces the separation, not just the screen.

Registration · Scheduling · Technologist · Radiologist (author) · Release (second reader) · Referring physicians · Clinical read access · Document management · System administration · Security audit · Backup

05 Security and traceability

Patient data does not leave your building. Who did what can be proven.

- Fully on-premises; the system needs no internet connection in daily use.
- Sign-in with a second factor for authors, releasers and administrators; read-only connection to LDAP or Active Directory.
- Access separated by site and department at database level.
- Tamper-evident audit trail: every change and every record access is kept in a chained series of checksums that can be recomputed at any time.
- Released reports and document versions are immutable; corrections are made as an addendum or a new version.
- Backup with recovery to any point in time; the procedure is tested automatically.
- Encrypted connections, security headers and request-rate limiting by the reverse proxy in front.
- Uploads are scanned for malware before storage; without a reachable scanner nothing is accepted.

06 Technology

Proven open building blocks, cleanly connected. No third-party licence fees for the core components.

Area	Implementation
Workplace	Web application for current versions of Chrome and Edge; nothing to install on the workstation
Server	Linux with Docker; all services as hardened containers without root privileges
Database	PostgreSQL 17 with row-level access separation and continuous backup
Image archive	Orthanc (DICOM C-STORE, DICOMweb, modality worklist)
Viewer	OHIF, integrated through short-lived access links
Sign-in	Keycloak (OpenID Connect, TOTP), optional LDAP / Active Directory
Interfaces	DICOM, DICOMweb, REST API with OpenAPI 3.1 description, Prometheus metrics
Not included	HL7/FHIR interfaces, billing, teleradiology, personal qualified signature
Server guideline	For a small department: 8 cores, 32 GB RAM, SSD for the database, mirrored image storage sized by study volume, UPS, separate backup medium. Exact sizing follows from the site survey.

07 Tested, not just claimed

All checks run automatically and are recorded in the project log. The figures come from the development machine and are no promise for your hardware.

Check	Result
Automated tests	More than 300 unit tests and 60 tests against a real database on every change
Whole workflow in the browser	Sign-in → appointment → examination → report → release by the second reader, with second factor, against a real server
Modality connection	Receive and worklist against a real DICOM archive with a simulated device, including Arabic names
30-minute endurance load	276,205 requests without an error, response time steady at 11 ms (95th percentile), memory stable
Recovery	11 GB database restored to a chosen point in time in 75 seconds after a simulated total loss
Report PDF	PDF/A-2b with embedded fonts, Arabic and German

08 From first talk to daily operation

We deliver the system as a finished installation: hardware, installation, training and support.

1. Demonstration and scope: which modules, how many workplaces, which modalities.
2. Site survey: device list, network, storage need, retention periods, users and roles.
3. Quotation covering hardware, installation and support.
4. Set-up and installation with a documented acceptance test, connection of every device.
5. Training per role with illustrated guides in Arabic and German.
6. Trial operation with your department, your approval, then support and updates.

09 What is included – and what is done on site

Included

- Software with all modules listed above
- Installation and operating manual, downtime procedure
- Training guides per role (Arabic, German) and training plan
- Interface description (OpenAPI)

Done at your site

- Connection and testing with your modalities
- Acceptance test on your hardware, including measured recovery time
- Independent security test if you require one
- Clarifying the legal requirements of your country

- Templates for site survey, acceptance and handover
- The content of your report templates

10 Frequent questions

Do we need internet?
No. The system runs entirely in your own network. Internet is only needed for remote support, if you want it.

May the viewer be used for diagnosis?
The built-in viewer is meant for review and is not released for primary diagnosis. For diagnosis you use an approved diagnostic workstation; the archive provides it with the images through DICOM.

Which modalities can be connected?
Devices that send DICOM. For the worklist at the device, the device must support modality worklist and be licensed for it. We clarify that in the site survey.

Is the system approved as a medical device?
No. Whether and which approval is required depends on your country and the intended use. We clarify this together before an installation.

What does it cost?
The price depends on hardware, number of devices and level of support. You receive a fixed quotation after the site survey.

Who owns the data?
You do. It lives on your server in open formats (DICOM, PostgreSQL, PDF/A).

See it for yourself. We show you the system with sample data and discuss what your department needs.
code-craft.tech/kontakt

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